# Weather's Impact on NYC Motor Vehicle Collisions Across Boroughs

*A Big Data ETL Pipeline Using MongoDB, Python, PostgreSQL, and Flask*



**APAN 5400**
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# Project Overview

**Objective:**

*Build an end-to-end Big Data analytics system to explore relationships between NYC vehicle collisions and weather patterns that is capable of supporting analytics and future scaling.*

**Technologies used**:

- **MongoDB** for raw ingestion
- **Python + pandas** for cleaning and joining
- **PostgreSQL** for warehousing
- **Flask** for API + Web Interface

Output: **Live dashboard** displaying insights & API endpoints

.



# Background

- Motor vehicle traffic collisions are leading cause of injury-related death for New York City (NYC) residents.[1]
- Adverse weather such as rain, snow, and fog reduces visibility and road traction, which can impact drivers' reactions to changing conditions.[2]
- Previous studies have shown that extreme temperatures have an influence on the risk level of road traffic accidents.[3]




1. Shaaban, Khaled. "Analysis and Identification of Contributing Factors of Traffic Collisions in New York City." *Transportation Research Procedia*, vol. 55, 2021, p. 1047. ScienceDirect, https://doi.org/10.1016/j.trpro.2021.07.145.

2.Romanowska, Anna. "Investigating the Impact of Weather Conditions and Time of Day on Traffic Flow Characteristics and Highway Performance." *Weather, Climate, and Society*, vol. 14, no. 3, 2022, pp. 467–478. American Meteorological Society, https://doi.org/10.1175/WCAS-D-22-0012.1.

3.Zou, Yajie, et al. "Exploring the Impact of Climate and Extreme Weather on Fatal Traffic Accidents." *Sustainability*, vol. 13, no. 1, 2021, p. 390. MDPI, https://doi.org/10.3390/su13010390.

# Business Case

- Evaluate whether weather conditions (precipitation, temperature, snow) significantly affect the number of traffic collisions across NYC boroughs
- Identify seasonal and weather-related patterns in collision frequency and severity, and examined geographic trends across NYC boroughs to uncover location-specific risk factors.
- Based on analysis findings, recommend future policies and countermeasures that will reduce crash occurrences.

# Audience

- **NYC Department of Transportation:** to improve overall transportation infrastructure and prevent future weather-related traffic collisions
- **NYC Emergency Management**: to improve emergency response to extreme weather situations



# ETL Pipeline Overview

NYC Open Data NOAA Weather APIs




MongoDB (Raw Data Layer)

Python + pandas (Transform Layer)




PostgreSQL (Data Warehouse)




Flask API Layer

**Front-End Dashboard**

MongoDB handles semi-structured JSON naturally, making it ideal for storing API responses without needing a strict schema. It allows us to ingest the raw data exactly as received, with no transformation required, which helps maintain reproducibility.

Our dataset fits in-memory so pandas is sufficient and provides powerful tools for:

- Cleaning null values
- Standardizing fields
- Converting data types
- Merging weather + collision datasets

PySpark didn't work because of a Java issue on computer

Strong relational database for our structured data that provides indexing, SQL querying, views, and joins.

Works with Flask.

Dockerized PostgreSQL ensured consistent environment for all team members.

Flask provides the interactive layer that turns our processed data into real-time API responses and dashboards.

# Overview of Data Source

- Source: NYC Motor Vehicle Collisions Crashes
  - Provided by NYC Police Department
  - Found on NYC OpenData
- Description: police-reported motor vehicle collisions in NYC
  - Mandatory report when someone is killed/injured or at least $1000 worth of damage
- Information: crash date, crash time, borough, zip code
- Time Range: 1/2021-12/2024

Source: Global Historical Climatology Network daily(GHCNd)

  - Provided by National Centers for Environmental Information (NCEI)
  - Found on NCEI government website
- Description: daily weather reported from the GHCNd station located in Central Park in NYC
- Information: daily min temp, max temp, rain/melted snow (in), snow/hail (in)
- Time Range: 1/2021-12/2024

# Data Source & Procurement

**1. NYC Open Data API — Vehicle Collisions**

- 402,362 records (2021–2024) retrieved
- Retrieved data in 50,000-record increments (API maximum)
- Automated loop to fetch all available data
- Stored in MongoDB as collection collisions

**2. NOAA / Weather API**

- **Challenge**: API restricts requests to 1-year maximum date range
- **Solution**: Created a function to split the 4-year timeframe into 48 monthly segments, then downloaded each month separately
- Retrieved daily weather metrics for 2021-2024
- Converted units to standard measurements and reorganized by date
- Stored in MongoDB as collection weather

**3. Kaggle - GeoJSON file of NYC zip codes**



# Data Quality & Licensing Considerations

**Data Quality Dimensions:**

- **Accuracy:** Validated numeric fields, removed corrupted rows
- **Completeness:** Handled missing boroughs/dates; weather gaps filled with NA
- **Timeliness:** Data updated through APIs with timestamp fields
- **Consistency:** Standardized date formats & weather metrics

**Licensing:**

- NYC Open Data is *public domain*
- NOAA weather data is open & free for reuse

**Scalability:**

- MongoDB handles large datasets well
- ETL can scale using batching
- API can scale horizontally (containerized deployment)



# LIVE DEMO!

# Associated Costs

- More cities/dates will grow our required storage needs as we scale up our selection of features
- MongoDB is already equipped with build in Horizontal Scaling (Sharding) if multiple servers are required for storage




- Advanced queries on large datasets may require large amounts of processing power for computations
- Migrate to cloud computing solutions for both storage and processing to expedite indexing and retrieval of datasets



# Conclusion & Future Recommendations

- Use the dashboard to create predictive models to identify when extreme NYC weather preparation plans likely need to be implemented
- Run further Pearson statistical tests and see if there is a significant difference in collision occurrences under different weather conditions.
- Future Directions: Expand the dashboard to include collisions/weather databases of other NY cities or counties



# Contributions to Project




API connection and mongoDB storage:

**Lauren Tran**




Data cleaning and transformation:

**Pamela Herrera**




Main data warehouse for easy connection to Flask:

**Cole Reimann**




Everyone contributed to this:

- Live API - **Pamela Herrera**
- Average Daily Collisions by Borough graph : **Lauren Tran**
- Heat map: **Cole Reimann**